Satellite Flood Mapping and Road Access in Mountain Valleys

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***Abstract*— Mountain floods leave water and deposited debris along the valley floor, while cloud and radar shadow prevent parts of the valley from being observed. Sentinel-1 backscatter change, Sentinel-2 spectral change and water segmentation are combined on a 10 m grid, with slope and height above drainage restricting detections on the valley sides. The flood extent is compared with pre-event OpenStreetMap buildings, roads and bridges, and affected road segments are removed before settlement road access is calculated. For the Bhote Koshi–Trishuli flood of 26 August 2026, the extent has F1 0.906 against four Copernicus EMSR927 areas, and the road-access calculation identifies 10,788 buildings without a road connection. Trishuli was used to select settings during development. The 27-event evaluation gives mean F1 0.546, whereas the three test events excluded from parameter selection have mean F1 0.366. Excluding snow from optical date selection changes the twelve-event development mean from 0.5335 to 0.5351; two additional flood cases have unchanged results.**

***Index Terms*—Flood extent, Sentinel-1, Sentinel-2, debris, road access, mountain valleys.**

I. INTRODUCTION

Flood water can recede before a clear satellite image is acquired, leaving debris on the valley floor that is absent from an open-water segmentation. The debris can increase radar backscatter, whereas smooth water often decreases it, so a decrease in backscatter alone does not describe the affected ground. Cloud removes optical observations and steep valley sides also cause radar shadow and layover. These conditions require the measured changes to be considered with the terrain and with the ground that remains unobserved.

We compare satellite images acquired before and after the event and combine the change detections with radar and optical water predictions. Height above drainage and channel connections limit detections on the slopes; on large low plains, where water may already be present before the event, the water predictions are used without requiring additional change. Buildings, roads and bridges are taken from pre-event OpenStreetMap. Removing affected segments from the road network then allows a settlement outside the mapped flood extent to be identified when its road connection has been lost. The Bhote Koshi–Trishuli flood is used as the case study, with further comparisons on mountain and lowland events.

II. DATA AND IMAGE PROCESSING

A. Input data

Sentinel-1 GRD is calibrated and terrain corrected from the ESA annotations (Table 1). The OpenStreetMap snapshot must precede the event, including the wider network used for road access; its replication time is checked before buildings and roads are read. Copernicus EMS and UNOSAT geometries are excluded from the mapping inputs and read separately for evaluation.

Table 1: Input data and processing.

Input	Processing and use
Sentinel-1 GRD, IW VV/VH	Backscatter calibration, noise removal and Range–Doppler terrain correction using Copernicus DEM and the EGM2008 geoid. Same-track images provide radar change and water-model inputs.
Sentinel-2 L2A	Cloud, cloud-shadow and snow masking; pre-event, first clear post-event and later reflectance composites.
Copernicus DEM GLO-30	Elevation, slope, D8 drainage, flow accumulation, height above drainage and walking times.
Pre-event OpenStreetMap	Buildings, roads, bridges, places and health facilities from a dated Geofabrik extract; historical Overpass or ohsome queries are fallbacks.
Kuro Siwo and Sen1Floods11	Published training splits for water-model weights and validation splits for checkpoint selection [1, 2]. The optical encoder uses ImageNet initialisation, as permitted by the organisers.

B. Radar observations

Radar images are processed within each orbit track because a change in viewing direction also changes the backscatter and the visible part of the valley. Four pre-event acquisitions provide the earlier observations. Calibration and noise removal use the ESA annotation tables, and Range–Doppler terrain correction uses Copernicus DEM with EGM2008 heights. Both increases and decreases in VV and VH are tested. A two-polarisation T^2 statistic uses moderated variance, with its change distribution estimated from pre-event pairs within each terrain group and a nominal false-alarm rate of 1% per track. The first post-event acquisition and the mean of the first two acquisitions are tested with Bonferroni correction; tracks with fewer than three pre-event acquisitions use the simpler change rule.

C. Optical observations

The Sentinel-2 scene-classification classes 2, 4, 5, 6 and 7 are retained when date coverage is calculated and reflectance is pro-

cessed. Snow, class 11, is excluded from both operations. The pre-event composite uses the five clearest acquisitions, and the first post-event composite takes the first valid observation at each pixel within twelve calendar days of the flood. Different pixels can therefore use different dates. A later composite uses the median of the four clearest post-event acquisitions.

Vegetation loss is measured where pre-event NDVI is at least 0.2. A debris detection requires

$$\begin{aligned} &\Delta\mathrm{NDVI} < -0.15 \quad \text{and} \\ &(\Delta\mathrm{BSI} > 0 \text{ or } \Delta B > 0.02)\,, \end{aligned} \tag{1}$$

where Δ is the post-event value minus the pre-event value and B is brightness. New water requires positive MNDWI after the event and negative MNDWI before it. Debris change must remain visible in the first clear image and the later composite, while water in the first clear image is retained because it can recede before the later image.

Ground that was already bare cannot supply the same vegetation-loss measurement. For pre-event NDVI below 0.3, a separate detector combines changes in MNDWI, NDVI, BSI, NBR2 and spectral angle. It is calibrated on two halves of the pre-event composite at a nominal 1% false-alarm rate. The change thresholds are estimated for each scene, whereas the false-alarm rates and terrain limits are the same across events and were selected on the development comparisons.

III. FLOOD-EXTENT CALCULATION

A. Water segmentation

The radar water prediction is the mean of three U-Nets with ResNet-34 or ResNet-50 encoders trained from random weights. Two models use Kuro Siwo and the third also uses Sen1Floods11 radar chips reprocessed from GRD. The six input channels contain post-event VV and VH and the two preceding VV and VH images. A 7×7 Lee filter is applied before prediction, and batch-normalisation statistics are estimated from scene tiles with height above drainage below 80 m. Predictions from eight rotations and reflections are averaged on GPU; CPU runs omit this averaging. The maximum probability across orbit tracks is retained.

The optical U-Net uses an ImageNet-initialised ResNet-34 encoder and nine channels: B2, B3, B4, B8, B11 and B12, with MNDWI, NDWI and NDVI. Sen1Floods11 supplies 252 hand-labelled training chips. Dark-object subtraction is applied to the visible bands, with blur and illumination variation during training to represent the optical composites. Validation selected epoch 32 at water IoU 0.811. New water requires probability above 0.5 after the event and below 0.3 before it; predictions are made for both the first clear image and the later composite.

The training labels describe open water. Most mountain reference maps also contain deposited debris and flood traces identified in later high-resolution optical images, where little water may remain. The water models have no debris class, and the optical change detections account for much of this ground. Water segmentation gives higher scores on the lowland flood comparisons.

B. Terrain restrictions

Height above nearest drainage (HAND) and slope restrict the detections in Table 2. On mountain valley floors, radar water predictions require nearby measured change, and each connected detection must reach a channel draining more than $20\,\mathrm{km}^2$. Above the height limits, agreement between radar and optical change is retained, or agreement between two masks where it connects to the likely extent. Radar change and radar water predictions share their satellite inputs and are treated as one sensor source when road-break evidence is assessed.

Table 2: Rules for the likely flood extent.

Terrain	Detection rule
Mountain valleys	Radar change below HAND 40 m; optical change below 80 m; radar water probability > 0.8 below 40 m with nearby measured change. Connected detections touch drainage $> 20\,\mathrm{km}^2$.
Flat valley ground	Mean slope $< 6^\circ$ over 310 m. Two of radar change, optical change and radar water probability > 0.7 agree within 20 m. Detections touch drainage $\geq 5\,\mathrm{km}^2$.
Large low plains	Connected flat ground $\geq 100\,\mathrm{km}^2$ and HAND $< 10\,\mathrm{m}$. Radar water probability > 0.5 or optical new water; no additional change or river-connection requirement.

On large low plains, water can be present in the pre-event images and produce little subsequent change, while cultivation can change the optical response of fields. River positions from the 30 m elevation model are also unreliable on this ground. The direct water rule is used in these areas. Wide water bodies present before and after the flood are removed by morphological opening with an eight-pixel radius, but narrow channel water adjoining the affected extent remains. Weaker detections near drainage, including radar water probability above 0.3 and early optical change, form the possible extent. Ground without a usable radar or optical view is marked unobserved.

C. Flood level beneath cloud

Cloud-covered valley ground has no optical change observation, although neighbouring river reaches can still have radar or clear optical detections. Drainage is divided into 300 m reaches and a flood level is selected for each reach from 17 heights between 0 and 60 m above the channel. Detections and observed non-detections are grouped by height above drainage. Their likelihood is combined along the D8 river tree, with a penalty for differences between neighbouring levels, and dynamic programming selects the levels. Detection and false-alarm rates are fitted by hard expectation-maximisation. The false-alarm rate is bounded below by its observed value on ground with HAND at least 60 m, and a detector is removed from the fit when that value exceeds 15%. Confined-valley pixels below the estimated level are added where optical observations are absent and the likely map has no detection. These pixels retain an inferred label and do not establish strong evidence of a road break.

IV. INFRASTRUCTURE AND ROAD ACCESS

Buildings are counted as exposed when their representative point lies within 10 m of the likely extent. Roads are split at shared vertices and into segments at most 50 m long; a segment is classified as likely impassable when at least half its length, or

a continuous 50 m length, lies in the likely extent. Possible status requires more than 20% of the segment within either extent. Bridges are assessed from their approaches, up to 30 m beyond each end, because the span itself normally crosses water.

Road and trail graphs cover the mapped area and a 15 km buffer. Routes to towns, hospitals and main-road exits are calculated before and after the affected segments are removed, with roads outside the mapped area assumed open. Strong road-break evidence requires persistent optical change, radar change from two orbits, or overlapping radar and optical change. A separate calculation includes weaker breaks and uncertain bridge failures. Buildings exposed to the flood and buildings that have lost road access are counted separately.

For a settlement without road access, the remaining trails are checked for a connection to an open road and walking time is estimated from the elevation model. The pre-event route and the breaks on that route are recorded, and repair candidates are ranked by the number of buildings reconnected when the affected segments reopen. A supplied upstream point can also be traced along D8 drainage, with nearby settlements listed by height above the path. Water depth and arrival time are not calculated.

V. EVALUATION

A. Events and selection of settings

The twelve development events, including Trishuli, were used to select change thresholds and terrain limits (Table 6). Five validation events include three lowland floods that informed the plains rule. Five further lowland events were used to compare the radar and optical water combination, so their reported scores also include parameter selection. Thame, Buner and Charikar were excluded from further selection after scoring. The Charikar result uses a different plains configuration from the combined radar–optical rule, which limits its comparability with the other events.

Copernicus EMS and UNOSAT provide the event reference maps [3, 4]. The five lowland evaluation references use COSMO-SkyMed for Sahuka, Sentinel-2 for Larkana, TerraSAR-X for Kurigram and Jamalpur, and ICEYE for Sharaqpur. Kłodzko uses EMSR756 from WorldView-3, and Cantiano uses EMSR634. The three-event test references are UNOSAT 3919 for Thame, EMSR838 AOI10/11 for Buner and EMSR459 AOI01 for Charikar. The UNOSAT Trishuli reference shares the 27 August optical acquisition with the mapping inputs, and the Far-West and Terai references use Sentinel-1. Ahr shares a storm with the Kuro Siwo Belgium training and Netherlands validation events, approximately 85–120 km away, although the basins do not share labelled pixels.

B. Scoring and comparisons

Precision, recall, F1 and intersection over union (IoU) are calculated within the analysed extent declared by each reference product. For true-positive, false-positive and false-negative pixel counts TP, FP and FN,

$$P = \frac{TP}{TP+FP}, \qquad R = \frac{TP}{TP+FN}, \qquad F_1 = \frac{2TP}{2TP+FP+FN}, \quad \mathrm{IoU} = \frac{TP}{TP+FP+FN}. \tag{2}$$

The likely extent and the likely-plus-possible extent are scored separately, with equal weight for each event in an event mean. Trishuli's event score averages its EMSR927 and UNOSAT comparisons; the pooled EMSR927 score is reported separately. Area mapped outside a reference's analysed extent is recorded but cannot be assessed by its overlap score. Shovi and Arstanbap are scored within 1 km of the reference, which increases their measured precision.

Changes are compared on the same events and reference extents, with a separate calculation withholding Sentinel-2. Paired event differences are bootstrapped where intervals are reported. Displacing copies of the development reference maps by one 10 m pixel gives mean F1 0.872 against the undisplaced references, with event values from 0.74 to 0.97. Narrow traces lose more overlap under this displacement; the calculation measures reference sensitivity to displacement. The 27-event benchmark uses optical date coverage that includes snow, while the effect of excluding snow is evaluated separately in Section VI.C.

The baselines use the same events and reference extents. The UN-SPIDER practice [5] is implemented with a VH after/before ratio of dB values above 1.25 after 50 m smoothing, followed by permanent-water removal, slope below 5° and a minimum of eight connected pixels. Stable water from Sentinel-2 and Copernicus DEM replace JRC water and HydroSHEDS in this implementation. The Otsu baseline thresholds post-event VV dB on gentle slopes and retains water present after, but absent before, the event.

C. Water-model evaluation

Four radar variants were compared on the Kuro Siwo test split when choosing encoder initialisation, and two of those variants are included in the ensemble. Their reported test scores include this use of the test split. The third radar model and the ensemble have no separate Kuro Siwo test evaluation. The optical test and Bolivia splits were scored once with the checkpoint selected on validation, and Bolivia is absent from the training split.

VI. RESULTS

A. Comparison with published water models

The optical U-Net has water IoU 83.3% on the Sen1Floods11 test split, compared with 83.1% for Prithvi-EO-2.0 600M-TL [6], 81.89% for SegFormer-B5 [8] and 83.41% for Prithvi-CAFE [9] (Fig. 1). On Bolivia, which is absent from training, IoU is 79.6%, compared with 77.95% for Prithvi-100M [7], 75.82% for SegFormer-B5 and 81.37% for Prithvi-CAFE. The two radar U-Nets have Kuro Siwo flood F1 78.9% and 78.8%, whereas Bountos et al. [1] report 80.12% for UNet-ResNet50 and 78.70% for FloodViT. Published values retain the training and preprocessing of the cited studies; the model comparisons therefore include these differences. The radar test scores also include the model-selection use described in Section V.C.

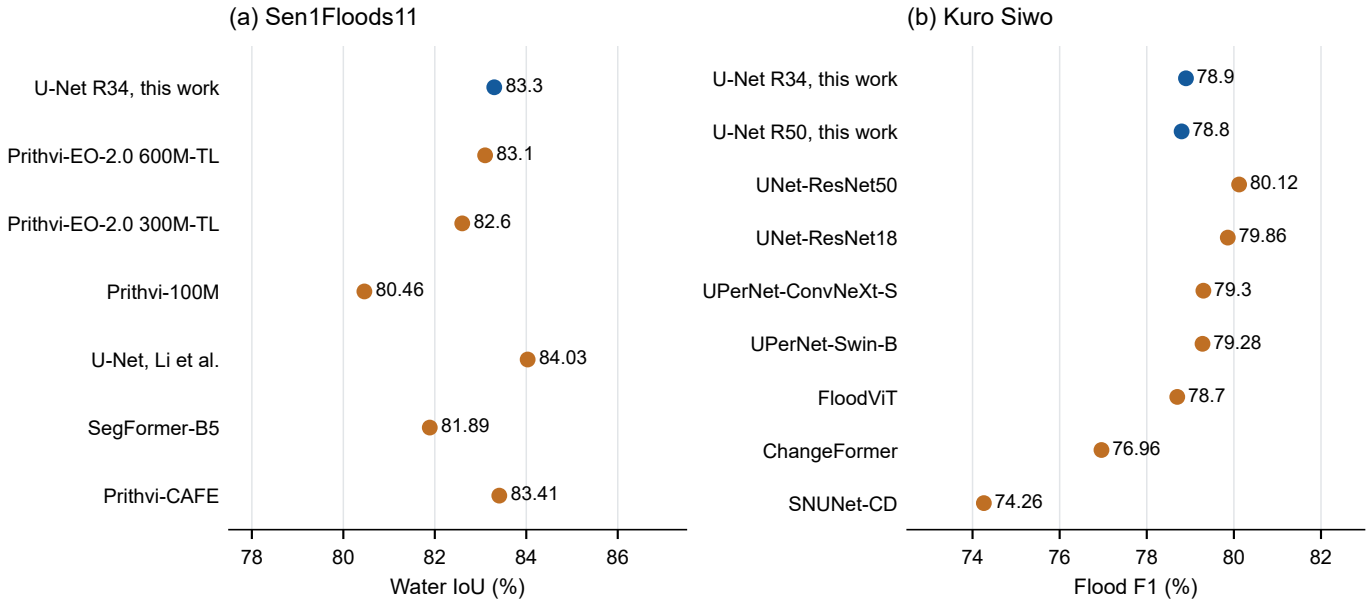


Figure 1: Water-model comparison on Sen1Floods11 (a) and Kuro Siwo (b). Blue points are this work; orange points are published model scores. Optical values are from Szwarcman et al. [6], the Prithvi-100M model card [7], Li et al. [8] and Kaushik et al. [9]. Radar comparisons are from Bountos et al. [1]. Published models were not rerun here. The two radar models from this work were compared on the test split during model selection.

B. Event mapping

The combined flood map has mean F1 0.546 across 27 events, compared with 0.258 for UN-SPIDER and 0.259 for Otsu thresholding (Fig. 2; Table 7). The combined map equals or exceeds all compared methods on 17 events. The radar–optical water union has mean F1 0.726 on the five lowland evaluation events, above the combined map's 0.704, while the mountain evaluation gives 0.508 at Kłodzko and 0.262 at Cantiano. Individual event scores are shown in Fig. 3.

The validation mean increases from 0.499 to 0.589 when the plains rule is added. On the three test events, mean F1 is 0.366: Thame 0.666, Buner 0.424 and Charikar 0.009. With Sentinel-2 withheld, the test mean is 0.294 (Fig. 4; Table 8). At Charikar, most of the 0.25 km^2 urban flood trace is missed, and the highest score among the tested methods is 0.04. Narrow mudflows and hillslope landslides also give low development scores at Arstanbap (0.146) and Jablanica (0.218).

C. Optical date selection

Snow-covered pixels contributed to the clear-cover fraction used to select Sentinel-2 dates, although snow was removed when reflectance composites were made. Excluding snow raises Kaski F1 from 0.5704 to 0.5906 and decreases Melamchi from 0.8558 to 0.8546. The other ten development events keep the same dates. Mean F1 changes from 0.5335 to 0.5351, with a paired 95% bootstrap interval for the change from −0.0003 to +0.0051; the interval includes zero. This comparison uses the development events and changes only the coverage calculation, with shadow masking held fixed.

The snow-exclusion comparison includes two additional flood areas (Table 3). Tzermiado uses EMSR933 AOI03 from GeoEye-1 and San Sebastiano uses EMSR800 AOI01 from Pléiades. Their selected dates and maps are unchanged, so these flood cases show no gain from the correction. At Tzermiado, recall is 0.905 but 10.50 km^2 is mapped within the analysed region against 3.76 km^2 in the reference, giving precision 0.324. Withholding Sentinel-2 reduces the extra mapped area and increases F1 to 0.548. The additional CPU runs omit prediction averaging over rotations and reflections.

Table 3: Snow exclusion in optical date selection.

Area	Snow included F1	Snow excluded F1	F1, no S2
Twelve development events	0.5335	0.5351	0.4168
Tzermiado 2026	0.4772	0.4772	0.5477
San Sebastiano da Po 2025	0.2347	0.2347	0.1020
Viksdalen 2024, landslide	0.0004	0.0004	0.0000

The Viksdalen EMSR775 AOI11 product was found to contain only one landslide polygon, with area 0.16 ha and thirteen cells on the 10 m grid. The landslide score is reported separately from the flood comparisons.

D. Water segmentation and added observations

The optical model has water IoU 0.833 and F1 0.909 on the Sen1Floods11 test split, and IoU 0.796 and F1 0.886 on Bolivia (Table 4). Adding optical new water to the combined map increases the twelve-event development mean F1 by 0.0066, with a paired 90% interval from 0.0001 to 0.0138. With Sentinel-2 withheld, estimated flood levels increase mean F1 from 0.371 to 0.417, a change of 0.046 with an interval from 0.023 to 0.073; ten events improve and none decline. Whole-map flood-level estimates score 0.47–0.49 against 0.527 for the combined map, so the estimated level is used only for unobserved valley ground.

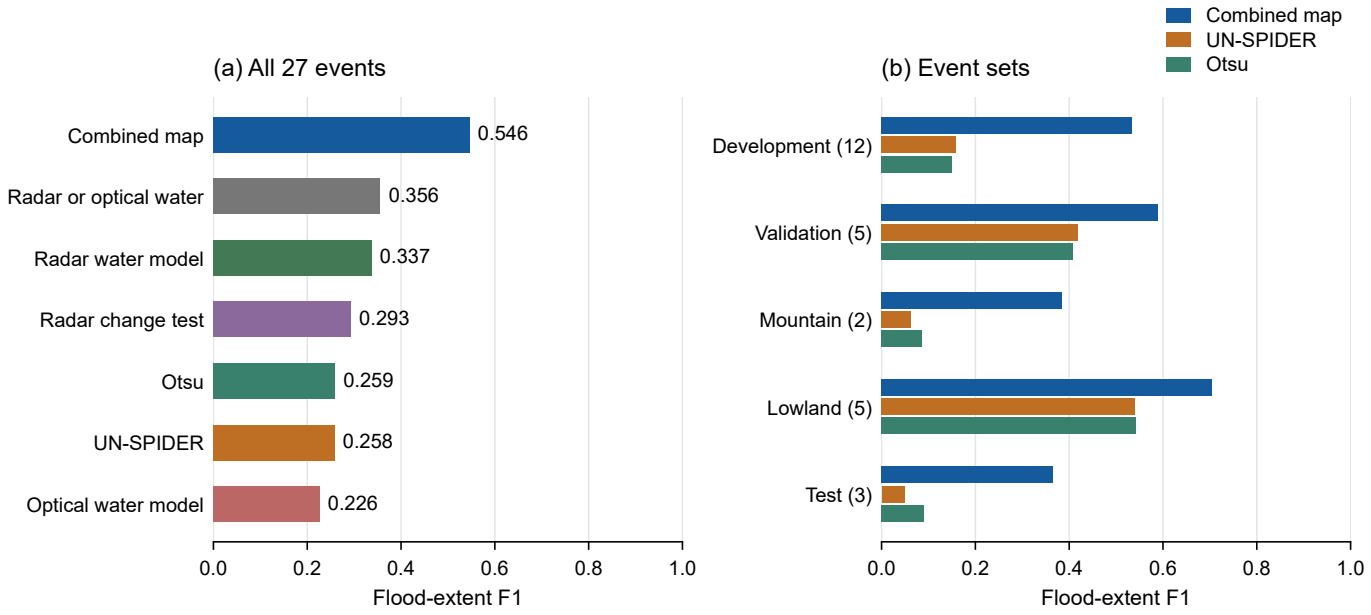


Figure 2: Mean flood-extent F1 by method (a) and event set (b). Events have equal weight. Development, validation and lowland evaluation scores include parameter selection.

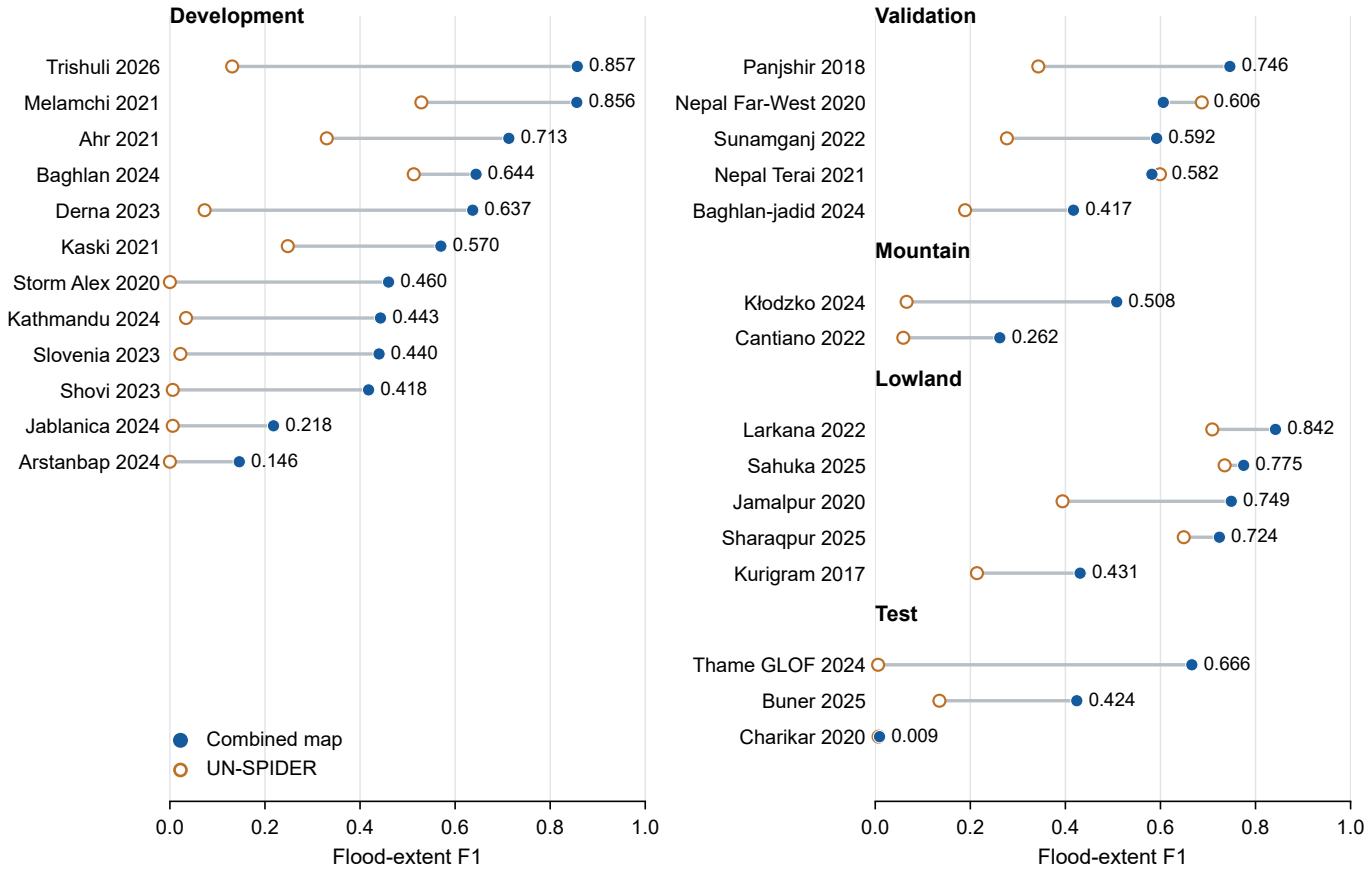


Figure 3: Flood-extent F1 for the combined map and UN-SPIDER. Labels are combined-map scores. The three test events were excluded from further parameter selection after scoring. The snow-exclusion experiment is reported separately in Section VI.C.

E. Trishuli case study

The Trishuli run covers 6,080 km^2 and starts from raw satellite data. Its likely extent has pooled F1 0.906 against EMSR927 AOIs 01, 02, 03 and 05, with precision 0.906 and recall 0.907 (Table 5, Fig. 5). The UNOSAT corridor comparison gives F1 0.815, precision 0.875 and recall 0.763. These are development results because Trishuli was used to select mapping settings. On the 49.5 km^2 where both references map the event, EMSR927 and UNOSAT agree at F1 0.938, while the combined

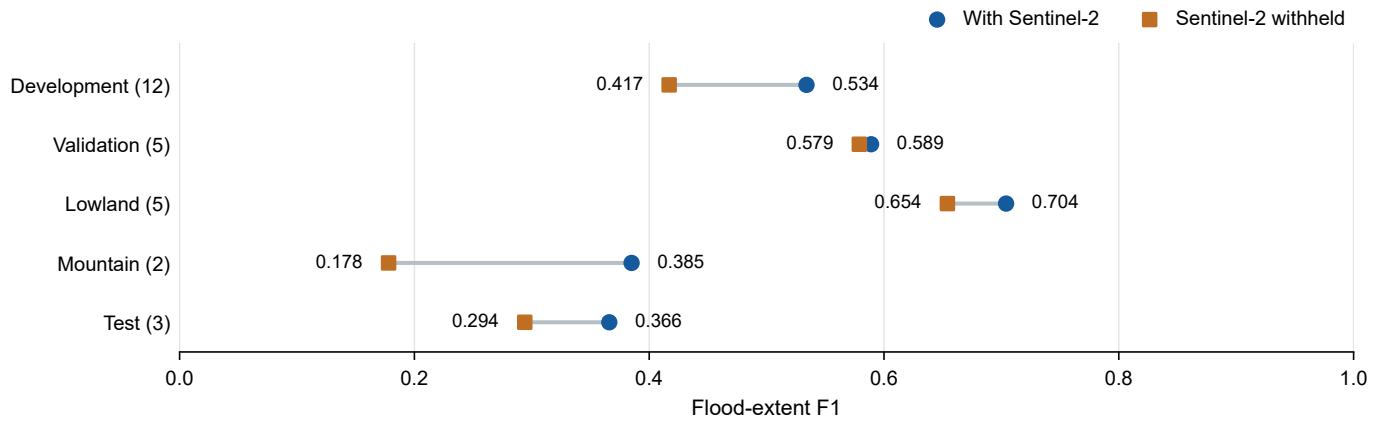


Figure 4: Mean flood-extent F1 with Sentinel-2 and with Sentinel-2 withheld. The event sets and reference extents are unchanged between the two calculations.

Table 4: Water-model test results. Kuro Siwo test scores were observed during development.

Model and test set	IoU	F1	Precision	Recall
Optical U-Net, Sen1Floods11	0.833	0.909	0.937	0.883
Optical U-Net, Bolivia	0.796	0.886	0.903	0.870
Radar ResNet-34, Kuro Siwo	—	0.789	—	—
Radar ResNet-50, Kuro Siwo	—	0.788	—	—

map scores 0.907 and 0.905 against them. UN-SPIDER scores 0.26 on this area and the radar water model alone 0.69. At Baghlan, the Pléiades and Sentinel-2 reference maps agree at 0.845.

Table 5: Trishuli comparison with EMSR927. Building and road comparisons use a 20 m tolerance.

Quantity	AOI01	AOI02	AOI03	AOI05
Flood-extent F1	0.912	0.862	0.915	0.904
Building recall	0.68	0.84	0.81	0.55
Building precision	0.95	0.85	0.89	0.72
Road-length recall	0.76	0.69	0.88	0.66
Road-length precision	0.81	0.73	0.83	0.71

The likely extent covers 72.1 km^2, or 153.1 km^2 when possible detections are included. There are 53.4 km^2 without observations and 4.6 km^2 estimated beneath cloud. The likely extent contains 3,958 buildings, with 126 km of motorable road and 60 motorable and 68 foot bridges affected. Road access is lost for 10,788 buildings in 58 named settlements, including 4,415 without a mapped trail avoiding the extent. Reopening 8.8 km of NH42 near Khangjim, including a bridge, reconnects 2,993 buildings in the road calculation.

Settlement access was also calculated using EMS-graded road and bridge breaks over the same road graph and EMS area. The buildings classified without road access agree at precision 0.874, recall 0.835 and F1 0.854 between the EMS-break and mapped-break calculations. Differences include 1,408 buildings behind breaks present only in the mapped extent and 789 behind EMS breaks classified as possible. The mapped infrastructure damage and road access have not been checked on the ground.

F. Outputs and runtime

The dashboard contains the extent, settlements without road access, their pre-event routes and the breaks along those routes (Fig. 6). English and Nepali situation reports use the same counts. Map data are embedded in one HTML file, while its display libraries load from the internet. Repair candidates and settlements along a supplied upstream flow path can be inspected with the map.

Raw-data runs on an eight-vCPU CPU machine take 19–46 min for areas of about 500–1,500 km^2, and 1 h 51 min for the 6,080 km^2 Trishuli box. The Sikkim GLOF of 4 October 2023 was run without selecting settings on that event and takes 36.5 min on an L4 GPU for 1,311 km^2; radar water prediction takes 3.3 min and optical water prediction 0.6 min. Its likely extent is 13.3 km^2, with 806 buildings without road access in Dzongu and a bridge near Sangkalang as the first repair candidate. The Sikkim outputs have no reference-map score.

VII. ABLATION STUDIES

Retraining radar water models with Himalayan non-flood samples reduces mean F1 by 0.103 on the three tested Himalayan events, with an interval from −0.155 to −0.050, so those weights are not used (Table 9). Relaxed tributary connections, revised vegetation-change detectors, supported early debris and Lee Sigma filtering before terrain correction also fail to improve the development mean sufficiently. Excluding snow gives the small development gain in Section VI.C, while leaving the two additional flood maps unchanged.

VIII. LIMITATIONS

The Trishuli overlap was measured on a development event, while the three test events have mean F1 0.366. Validation and lowland evaluation scores include parameter selection. Most of the narrow urban trace at Charikar was missed. Water segmentation also misses debris after water has receded, and river restrictions can exclude small mudflows and hillslope landslides. At Kurigram, water already present before the event produces little change and the mapped extent covers 29% of the reference flood area. The direct water rule improves some lowland comparisons but does not remove these omissions.

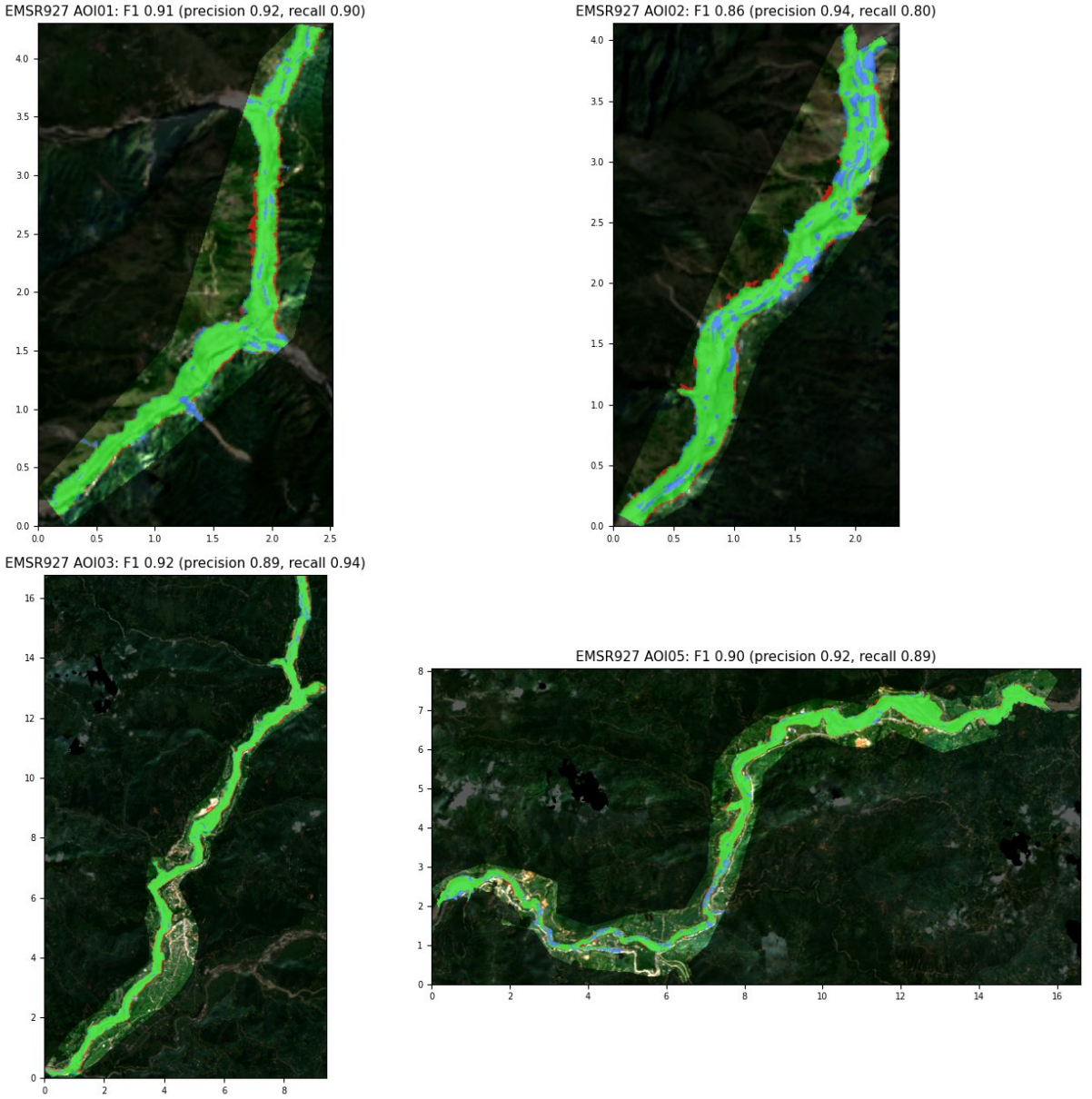


Figure 5: Trishuli likely extent and EMSR927 observed-event polygons. Green indicates overlap, red the mapped extent only, and blue the EMS reference only; ground outside each analysed area is dimmed. Background: post-event Sentinel-2 composite. Axes are in kilometres. Contains modified Copernicus Sentinel data 2026. Reference: European Union, Copernicus Emergency Management Service data.

The images used here are separated by 6–12 days, so the extent describes conditions at the acquisition dates and cannot warn of a glacier collapse minutes ahead. Where cloud prevents an optical observation, the extent added below the estimated river level is inferred from neighbouring observations. Exposure of a building to this extent does not establish destruction, and building counts do not establish population. Road access depends on pre-event OpenStreetMap coverage; Derna has few mapped buildings, and a route remaining in the graph has not been checked on the ground. Roads beyond the mapped area are assumed open.

Reference maps contain water, deposited debris or both, with different treatment of the river channel and acquisition dates. Some references share a sensor or an optical acquisition with the mapping inputs, while the five lowland evaluation references use other sensors. Areas outside the declared analysis extent have no reference assessment. The snow correction has a small development gain whose interval includes zero and leaves the two additional flood results unchanged; the uneven mountain-event results remain unresolved.

DATA ATTRIBUTION

Contains modified Copernicus Sentinel data 2026 and 2017–2025 for the evaluation events. Produced using Copernicus WorldDEM-30 © DLR e.V. 2010–2014 and © Airbus Defence and Space GmbH 2014–2018, provided under COPERNICUS by the European Union and ESA; all rights reserved. OpenStreetMap data: © OpenStreetMap contributors, with derived layers under ODbL. Satellite and DEM access uses Microsoft Planetary Computer, and dated OpenStreetMap extracts use Geofabrik. Training data are Kuro Siwo and Sen1Floods11. The

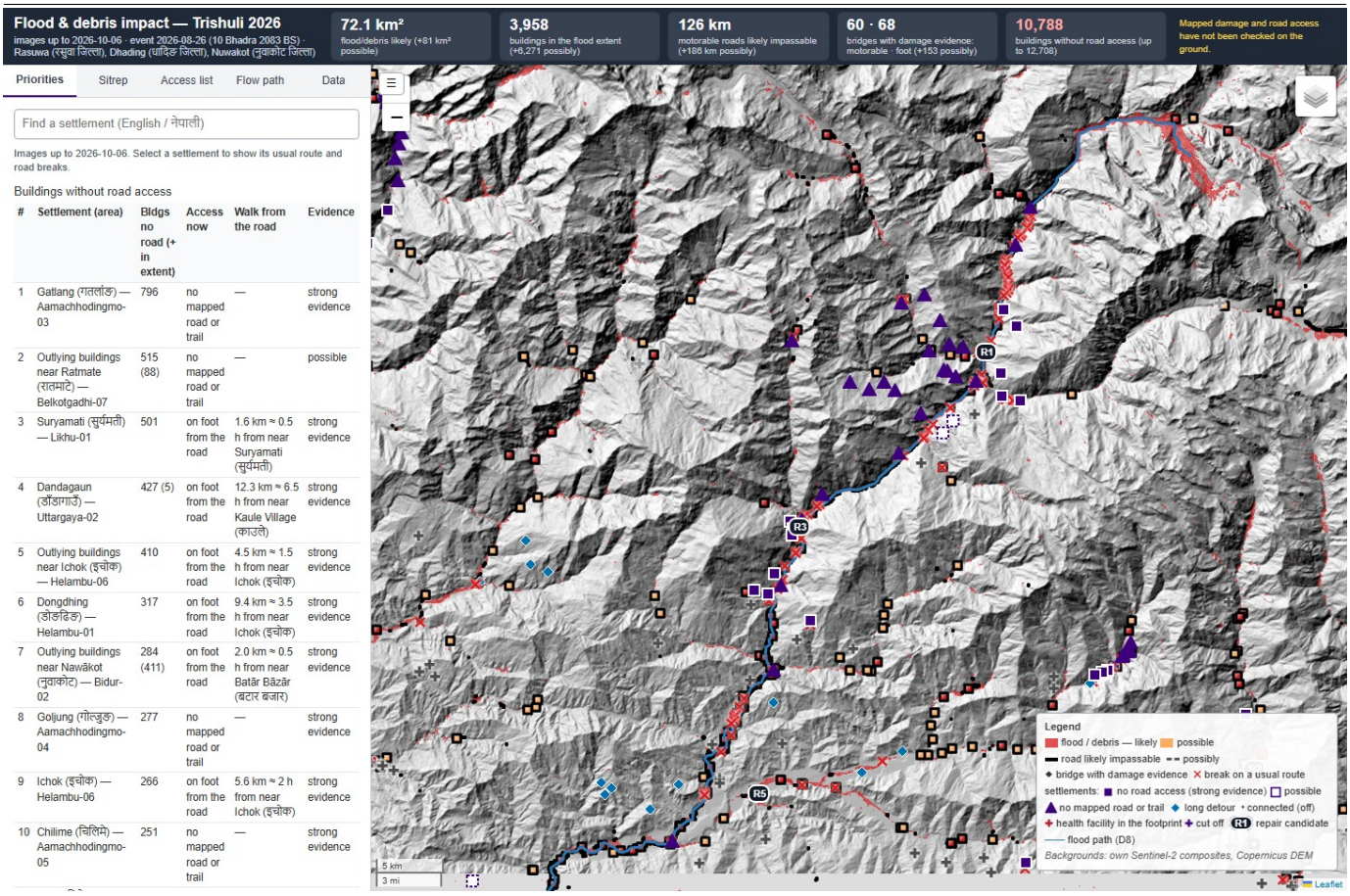


Figure 6: Trishuli dashboard. The settlement list gives road access, remaining trails, walking times and road-break evidence. Selecting a settlement displays its usual route and breaks. The map includes likely flood extent, settlements without road access, repair candidates and the drainage path from an upstream point. Background: Copernicus DEM hillshade.

optical encoder uses ImageNet-pretrained ResNet-34 through segmentation-models-pytorch. Copernicus EMS and UNOSAT products are used for evaluation only. Claude and Codex were used to edit code and documentation.

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A. EVALUATION TABLES

Table 6: Event sets and their use.

Set	Events	Use
Development (12)	Trishuli 2026, Melamchi 2021, Kaski 2021, Kathmandu 2024, Baghlan 2024, Shovi 2023, Arstanbap 2024, Storm Alex 2020, Slovenia 2023, Jablanica 2024, Ahr 2021, Derna 2023	Change thresholds, terrain limits, model settings and road-access rules selected on these events.
Validation (5)	Panjshir 2018, Baghlan-jadid 2024, Nepal Far-West 2020, Nepal Terai 2021, Sunamganj 2022	Three lowland events used to select the plains rule.
Lowland evaluation (5)	Sahuka 2025, Larkana 2022, Kurigram 2017, Sharaqpur 2025, Jamalpur 2020	Scores used to compare the combined radar–optical water rule.
Mountain evaluation (2)	Kłodzko 2024, Cantiano 2022	Evaluated once per method.
Test (3)	Thame GLOF 2024, Buner 2025, Charikar 2020	Excluded from parameter selection after scoring.
Additional floods (2)	Tzermiado 2026, San Sebastiano da Po 2025	Snow exclusion in optical date coverage evaluated with the remaining processing held fixed.
Additional landslide	Viksdalen 2024	Landslide result reported separately from the two additional flood cases.
Qualitative	Bhote Koshi 2025, Sikkim 2023	Visual or runtime checks without a scored reference comparison.

Table 7: Mean flood-extent F1 across 27 events.

Method	Dev. (12)	Validation (5)	Mountain (2)	Lowland (5)	Test (3)	All (27)
Combined map	0.534	0.589	0.385	0.704	0.366	0.546
UN-SPIDER	0.158	0.419	0.063	0.540	0.049	0.258
Otsu	0.151	0.409	0.086	0.542	0.090	0.259
Radar change test	0.258	0.445	0.074	0.383	0.176	0.293
Radar water model	0.225	0.535	0.029	0.667	0.115	0.337
Optical water model	0.206	0.166	0.021	0.515	0.066	0.226
Radar or optical water	0.238	0.539	0.029	0.726	0.126	0.356

Table 8: Mean F1 with Sentinel-2 and with Sentinel-2 withheld.

Events	With S2	S2 withheld
Development (12)	0.534	0.417
Validation (5)	0.589	0.579
Mountain evaluation (2)	0.385	0.178
Lowland evaluation (5)	0.704	0.654
Test (3)	0.366	0.294

Table 9: Ablation studies. Changes are paired differences in mean flood-extent F1.

Method change	Evaluation	Result
Himalayan non-flood samples for radar training	Three Himalayan events, HAND below 40 m	-0.103 [-0.155, -0.050].
Estimated flood level as the whole map	Twelve development events	F1 0.47–0.49; radar and optical change with radar water predictions gives 0.527.
Add optical new water	Twelve development events	$+0.0066$ [90% interval 0.0001, 0.0138].
Estimate extent beneath cloud	Twelve development events, S2 withheld	$+0.046$ [0.023, 0.073]; ten events improve and none decline.
Radar water without river-connection requirement on plains	Three lowland events and fourteen other events	Lowland changes $+0.09$ with S2 and $+0.19$ without S2; other-event mean $+0.001$.
Omit optical change on plains	Development events and Larkana	Development mean $+0.006$; Larkana -0.26.
Combine radar and optical water on plains	Development lowland events; Sharaqpur and Jamalpur	Development mean $+0.014$; two-event changes $+0.0135$ with S2 and $+0.052$ without S2.
River connections at 2 or 5 km^2	Twelve development events, four settings	Maximum mean change $+0.0022$.
Calibrated or recurring vegetation change	Twelve events, four settings; prior-year imagery available for eight	Mean changes -0.0025 to $+0.0009$.
Early debris with radar or model support	Twelve development events, two settings	Mean changes -0.0010 and -0.0017.
Lee Sigma before terrain correction	Twelve development events	Mean changes -0.0006 with S2 and -0.0008 without S2.
Exclude snow from optical date coverage	Twelve development events and two additional flood areas	Development mean $+0.0016$; additional flood maps unchanged.

Table 10: Individual-mask F1. Development and test columns are restricted below HAND 40 m; the other columns use their reference extents. The radar change rule differs from the calibrated radar change test in Table 7.

Mask	Development (12)	Validation (5)	Lowland (5)	Test (3)
Radar change rule	0.323	0.428	0.427	—
Radar water model	0.323	0.535	0.667	0.190
Optical water model	0.227	0.166	0.400	0.076
Radar or optical water	0.350	0.539	0.716	0.224

Table 11: Combined-map F1 by event. Snow exclusion in the date-selection coverage changes Kaski to 0.5906 and Melamchi to 0.8546 (Section VI.C).

Set	Event	F1
Development	Trishuli 2026	0.857
	Melamchi 2021	0.856
	Ahr 2021	0.713
	Baghlan 2024	0.644
	Derna 2023	0.637
	Kaski 2021	0.570
	Storm Alex 2020	0.460
	Kathmandu 2024	0.443
	Slovenia 2023	0.440
	Shovi 2023	0.418
	Jablanica 2024	0.218
	Arstanbap 2024	0.146
Validation	Panjshir 2018	0.746
	Baghlan-jadid 2024	0.417
	Nepal Far-West 2020	0.606
	Sunamganj 2022	0.592
	Nepal Terai 2021	0.582
Mountain evaluation	Kłodzko 2024	0.508
	Cantiano 2022	0.262
Lowland evaluation	Sahuka 2025	0.775
	Larkana 2022	0.842
	Kurigram 2017	0.431
	Sharaqpur 2025	0.724
	Jamalpur 2020	0.749
Test	Thame GLOF 2024	0.666
	Buner 2025	0.424
	Charikar 2020	0.009

Table 12: Published optical water IoU on the Sen1Floods11 hand-labelled test split, in percent. Bolivia is absent from the training split. Values are reported by the cited sources rather than rerun here.

Model	Test IoU (%)	Bolivia IoU (%)
Optical U-Net, this work	83.3	79.6
Prithvi-EO-2.0 600M-TL [6]	83.1	—
Prithvi-EO-2.0 300M-TL [6]	82.6	—
Prithvi-100M model card [7]	80.46	77.95
U-Net from scratch [8]	84.03	70.57
SegFormer-B5 [8]	81.89	75.82
Prithvi-CAFE [9]	83.41	81.37

Table 13: Kuro Siwo GRD flood F1, in percent. Published comparisons are from Bountos et al. [1], Table 2. The two radar members from this work were compared on this test split during development.

Model	Flood F1 (%)
ResNet-34 radar U-Net, this work	78.9
ResNet-50 radar U-Net, this work	78.8
UNet-ResNet50, best reported	80.12
UNet-ResNet18	79.86
UPerNet-ConvNeXt-S	79.30
UPerNet-Swin-B	79.28
FloodViT	78.70
ChangeFormer	76.96
SNUNet-CD	74.26